LAB 5: CONFIGURATION OF STATIC ROUTES AND DEFAULT ROUTES

OBJECTIVES:

1. To understand the concept and working principle of Static Routing and Default Routing in a computer network.
2. To implement a network topology using Cisco Packet Tracer and establish communication between two different LAN networks using manual routing techniques.
3. To analyze how routers forward packets based on routing table entries.

SOFTWARE AND HARDWARE REQUIREMENTS:

- Cisco Packet Tracer (Version 6.2 or higher)
- Windows PC / Laptop

THEORY:

1. Static Routing

Static routing is a manual method of configuring routing information in a router. In this approach, the network administrator defines the exact path that packets must follow to reach a destination network. Unlike dynamic routing protocols, static routing does not automatically adjust to network changes.

In static routing, entries are manually added to the routing table of a router. When a packet arrives, the router checks its routing table and forwards the packet based on the manually configured route. If the network topology changes or a link fails, the route must be manually updated by the administrator.

Static routing is commonly used in small-scale networks where the topology remains stable and does not change frequently. It provides greater security and administrative control because only predefined routes are allowed.

Operation:
It works by using the command:

```
ip route [destination_network] [subnet_mask] [next_hop_address]
```

The router looks at this manual entry to determine where to forward the packet. If the link fails, the administrator must manually update the route.

Application:
It is used in small networks where the topology is simple and does not change frequently. It is also used for security purposes to hide parts of a network.

2. Default Routing

Default routing is a special type of static routing that acts as a fallback path. It is also referred to as the “gateway of last resort.” If a router receives a packet whose destination is not present in its routing table, the router forwards that packet using the default route.

Default routing is especially useful in stub networks, where there is only one possible exit path to another network or the internet. Instead of maintaining multiple routing entries, a single default route simplifies the routing table and reduces administrative complexity.

This method ensures that unknown or external traffic is automatically forwarded to a predefined next-hop router.

Operation:
It involves configuring a route with all zeros:

```
ip route 0.0.0.0 0.0.0.0 [next_hop_address]
```

This tells the router to send any unknown traffic to the specified neighbor.

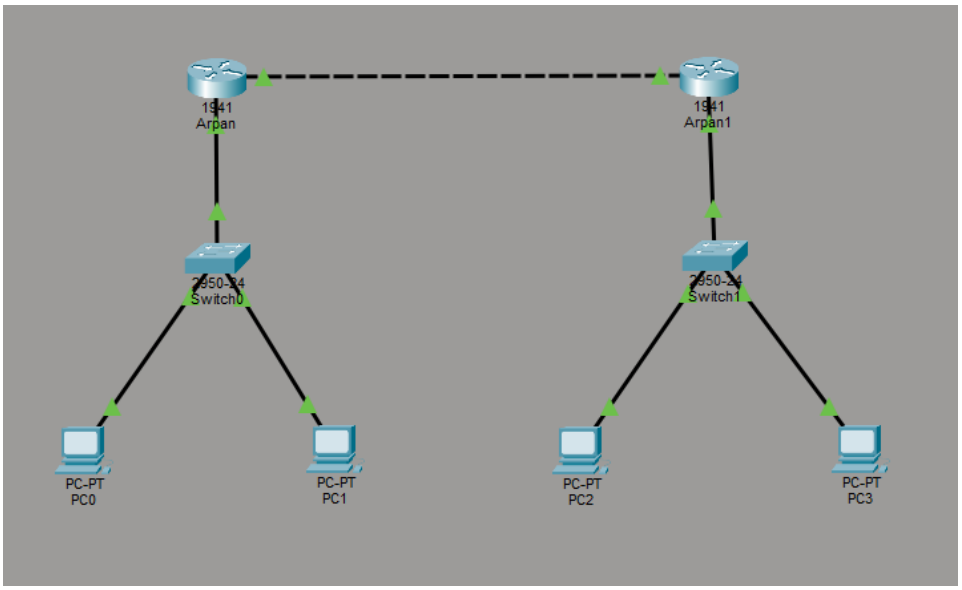
Application:
It is mainly used in stub networks (networks with only one exit path) or to connect a network to the Internet via an ISP. It reduces the size of the routing table.

NETWORK TOPOLOGY:

A topology was created using two routers: **“Arpan”** and **“Router1.”** These routers are connected via a serial link.

- Switch0 connects PC0 and PC1 to the **“Arpan”** router, forming the first network.
- Switch1 connects PC2 and PC3 to **“Router1,”** forming the second network.

This setup allows communication between two separate LAN networks through routing configuration.



CONFIGURATION TABLE:

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
Router Arpan	GigabitEthernet0/0	10.0.0.1	255.0.0.0	N/A
Router Arpan	GigabitEthernet0/1	192.168.1.1	255.255.255.0	N/A
Router1	GigabitEthernet0/0	10.0.0.2	255.0.0.0	N/A
Router1	GigabitEthernet0/1	192.168.2.1	255.255.255.0	N/A
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1

ROUTING CONFIGURATION:

Device	Routing Type	Command / Route
Router Arpan	Static Route	`ip route 192.168.2.0 255.255.255.0 10.0.0.2`
Router1	Static Route	`ip route 192.168.1.0 255.255.255.0 10.0.0.1`
Router Arpan	Default Route (Alt)	`ip route 0.0.0.0 0.0.0.0 10.0.0.2`

RESULT:

The network was successfully configured. To verify the connectivity, a ping request was sent from **PC1 (192.168.1.3)** in the first network to **PC2 (192.168.2.3)** in the second network.

The packets were sent and received successfully, confirming that the static route was correctly configured. Similar testing was performed for the default route configuration, and the connectivity was verified successfully with no packet loss.

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=10ms TTL=126
Reply from 192.168.2.3: bytes=32 time=11ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 5ms
```

DISCUSSION AND CONCLUSION:

During the lab session, the concepts of static routing and default routing were implemented to understand how routers forward packets between different networks. By manually configuring static routes, we gained full control over the path taken by data packets. This provided better understanding of routing table management and network control.

The default route was configured to act as a gateway of last resort. This simplified the routing table and demonstrated how unknown traffic can be forwarded efficiently through a single defined path.

Through successful packet transmission between two different LAN networks, it was confirmed that both static and default routing were correctly implemented in Cisco Packet Tracer.

Hence, the lab was completed successfully with proper understanding and practical implementation of static routing and default routing.